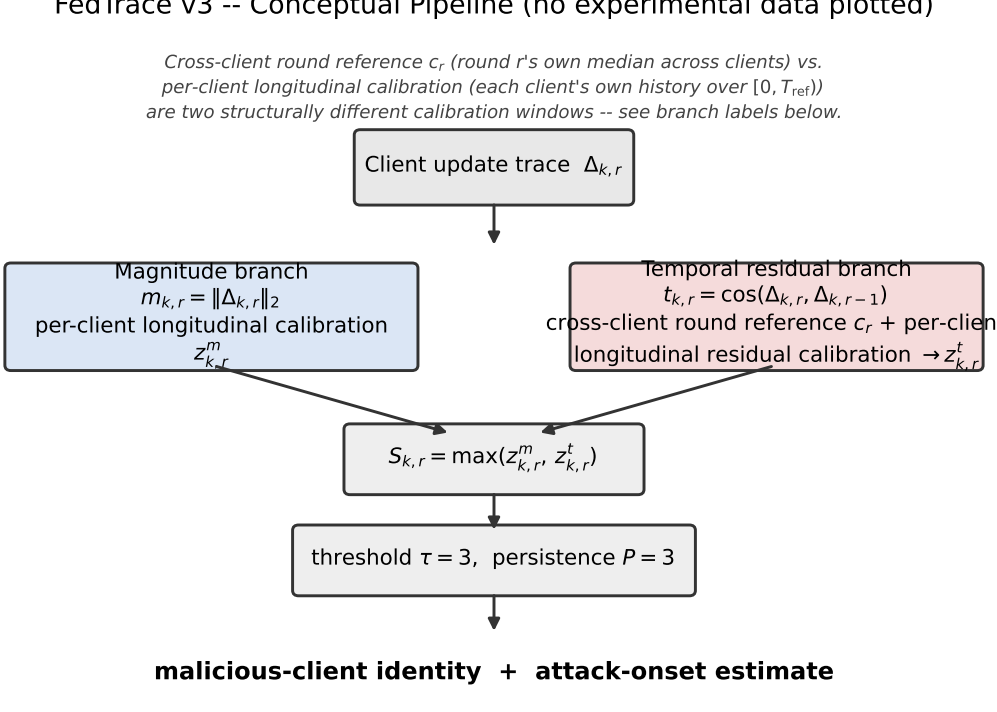
FedTrace v3 -- Conceptual Pipeline (no experimental data plotted)
Cross-client round reference c_r (round r's own median across clients) vs.
per-client longitudinal calibration (each client's own history over $[0, T_{\text{ref}})$)
are two structurally different calibration windows -- see branch labels below.
Client update trace $\Delta_{k,r}$
Magnitude branch
$m_{k,r} = \|\Delta_{k,r}\|_2$
per-client longitudinal calibration
$z^m_{k,r}$
Temporal residual branch
$t_{k,r} = \cos(\Delta_{k,r}, \Delta_{k,r-1})$
cross-client round reference c_r + per-client
longitudinal residual calibration $\rightarrow z^t_{k,r}$
$S_{k,r} = \max(z^m_{k,r}, z^t_{k,r})$
threshold $\tau = 3$, persistence $P = 3$
malicious-client identity + attack-onset estimate